

Headache

with TOBY HALL

In the beginner's mind there are many possibilities, in the expert's mind there are few.

Shunryu Suzuki, *Zen Mind, Beginner's Mind*

Headache is one of the most prevalent pain conditions present at all stages of life. It affects the majority of the global population and is among the top 10 causes of disability.¹ Despite this, the burden of headache has been largely underestimated, with headache disorders often dismissed as minor and undeserving of medical care.^{1, 2} Although athletes suffer from the same causes of headache as non-athletes, there are several key causes of headache that relate directly to exercise or to the sport itself. An additional factor to consider is that headache can impair both training and sports performance.

The International Headache Society (IHS) classifies headache into a range of disorders comprising three broad categories with more than 300 underlying headache forms.³ The majority of these disorders are not particularly relevant to an athletic population or the IHS definitions do not adequately cover the symptoms experienced by athletes.⁴ Figure 21.1 lists the broad classification according to the IHS and the common headache disorders seen in sport.

HEADACHE IN SPORT

The prevalence of headache directly related to sport has been poorly investigated or reported, although there is increasing attention in contact sport on concussion or trauma-related headache.⁵ In a New Zealand university setting, 35% of athletes surveyed reported headaches, both traumatic and non-traumatic.⁶ Importantly, 54% of people who reported headache associated with sport or exercise gave up their activity.⁷ In some codes of football, more than 50% of athletes experience regular headaches from matches and training.⁴

While the latest edition of the IHS headache classification is comprehensive, it is not aimed specifically at a sporting

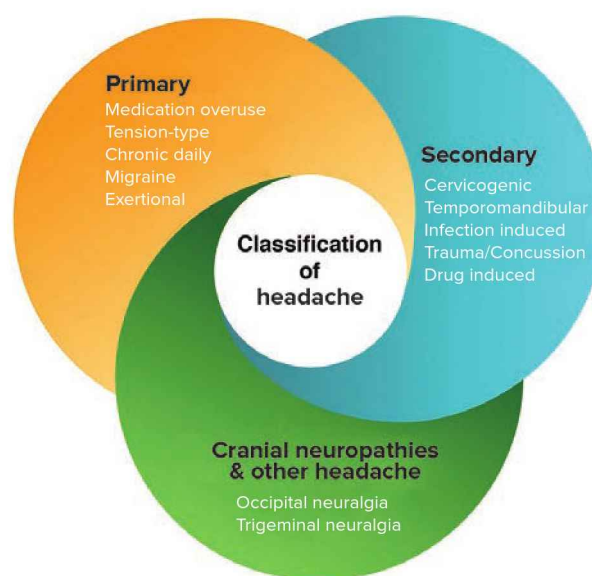


Figure 21.1 Classification of headache

setting. Williams and Nukada proposed a simple framework for specific sporting headaches.⁸ This included:

- effort-exertional headache
- effort migraine
- trauma-triggered migraine
- post-traumatic headache.

An alternative sports-specific headache organisational framework has been proposed that incorporates the non-sporting as well as sporting headaches.⁹ Headaches are categorised as follows:

- a recognised headache syndrome (migraine, tension-type headache, cluster headache) coincidental to sporting activity

- headache arising from mechanisms that occur during exertion; these can be primary (mechanism unknown) or secondary where a causal factor can be demonstrated, such as a headache:
 - related to changes in cardiovascular parameters
 - related to trauma
 - arising from structures in the neck
 - arising from mechanisms that are specific to an individual sport.

CLINICAL APPROACH TO THE PATIENT WITH HEADACHE

Headache is an unusual condition being both a symptom and a disease, which can make diagnosis and management challenging. Of primary importance is identifying when headache is a symptom of serious pathology such as a tumour, aneurysm, meningitis, subdural haemorrhage or space-occupying lesion. These are life-threatening events that the clinician must be vigilant for, and if suspected require urgent medical evaluation. Hence, a good knowledge of the following warning signals, or red flags, is essential:

- new, unaccustomed or atypical headache
- stiff neck or meningeal signs
- systemic symptoms (e.g. fever, weight loss, malaise)
- neurological symptoms (e.g. drowsiness, weakness, numbness of limbs)
- local extracranial symptoms (e.g. ear, sinus, teeth)
- changes in the pattern of headache
- headache increasing over a few days
- sudden onset of severe headache or headaches that wake the patient
- headache triggered by cough or Valsalva.

The majority of headaches are benign and do not require such intensive medical assessment.

PRACTICE PEARL

In each case, the clinician should seek a potential underlying cause for the headache, as this will influence management.

Common illnesses that may provoke headache include respiratory tract infection, sinusitis and influenza. Drug-induced headaches are also common. The most common are alcohol and caffeine, but many other commonly used drugs can also provoke headache including analgesics, antibiotics, antifungals and antihypertensives. Of particular importance is the frequency of analgesic use. Analgesic medication used on more than 15 days per month may

sensitise the trigeminocervical nucleus, and is defined as medication overuse headache.¹⁰ A thorough knowledge of the patient's drug history is therefore important.

Following the exclusion of headaches associated with red flags, drugs and infections, the clinician aims to differentiate between the more common headache categories: either primary or secondary headache. Primary headache includes migraine, exertional headache and tension-type headache. Secondary headache includes cervicogenic headache, temporomandibular dysfunction headache and concussion headache arising from musculoskeletal dysfunction around the neck and orofacial region.

Exertional headache follows exercise, and concussion post-traumatic headache follows a history of trauma. Tension-type headache typically presents without features of migraine or cervicogenic headache, and tends to be of a low-grade frequent bilateral headache described as pressing or squeezing. In contrast, differentiating cervicogenic headache from migraine is more challenging.

Although a comprehensive large-scale survey of people with 'pure' migraine and cervicogenic headache found that each has distinctive features (Table 21.1),^{11, 12} in many cases features coexist, which makes a definitive diagnosis difficult.

History

Headache location is unilateral in cervicogenic headache,¹³ but is less so in migraine, often starting on one side of the head and spreading to the opposite side during a migraine attack. The presence of associated neck and arm symptoms should also be noted. Cervicogenic headache usually starts in the neck before spreading to the head, whereas migraine starts in the head and spreads to the neck.

The nature of onset can be informative. A sudden onset of severe, unfamiliar headache, with altered consciousness or cognition, may indicate a cerebral haemorrhage or other intracerebral pathology.¹⁴ Cervicogenic headache typically has a more gradual onset. A recent history of a head and neck trauma, with or without concussion, is also relevant.

The intensity and temporal pattern is helpful in diagnosis. Migraine is episodic and more severe than cervicogenic headache, building up to a crescendo in a set time period (24–48 hours), unless early preventive analgesic medication is taken. Cervicogenic headache is typically less severe, does not have a set time period and does not progressively increase in intensity unless the provocative stimulus is maintained.

The relationship between exercise and headache onset is informative. Although exercise usually aggravates headaches of all types, brief episodes of headache brought

Table 21.1 Summary of subjective diagnostic criteria contrasting migraine and cervicogenic headache^{9, 10}

	Migraine	Cervicogenic headache
Gender ratio	1.7 female/male	0.7 female/male
Age at onset	18 years	33 years
Headache onset	Anterior head	Posterior head/neck
Pain area	50% unilateral	Predominantly unilateral
Nausea	Frequent	Infrequent
Photo/phonophobia	Very frequent	Infrequent
Throbbing pain	Frequent	Infrequent
Pain on trunk forward bend	Very frequent	Infrequent
Migraine medication	Usually helpful	Not helpful
Neck position pain	Rare	Universal

on by exercise may indicate exertional headache. More prolonged headache brought on by exercise might indicate cervicogenic headache or exertional migraine. Hence, a cyclist who develops headache while riding could have exertional headache, but equally if they have an inadequately set-up bike and associated poor posture this might be causing stress on the neck and hence cervicogenic headache. Once migraine has started, exercise is usually provocative. Conversely, aerobic exercise may be helpful in reducing the frequency of headache attacks.¹⁵

PRACTICE PEARL

Headache aggravated by specific neck movements or postures in sport may indicate cervicogenic headache, and identifying provocative movement is usually helpful for planning manual therapy.

Rest usually helps alleviate all headache forms during an attack. For example, migraine is usually relieved by sleep, but this might not always be the case for cervicogenic headache, particularly if the sleeping position, mattress or pillow is not suitable.

Prior to the onset of migraine headache, there may be associated visual or sensory symptoms (migraine with aura). Photophobia, phonophobia, nausea and vomiting are also commonly associated with migraine headache and usually occur concurrently with the headache episode. The presence of neurological symptoms or systemic symptoms such as weight loss and malaise is a 'red flag' and may indicate a more serious cause of headache (e.g. tumour).

Common forms of secondary headache are associated with symptoms in the upper or lower respiratory tract, sinuses and temporomandibular joints (jaw clicking or painful limitation of mouth opening), or influenza-like symptoms. These features should be looked for to enable differentiation from primary headache.

A past history of head trauma, even if relatively minor, may be significant as subdural haematoma may present well after the trauma, even weeks later. Previous problems such as encephalitis or major systemic illnesses should also be recorded.

Psychological comorbidities are factors in cases where headache becomes long lasting, such as in chronic migraine where headache persists for more than 3 months.¹⁶ An assessment of life stresses is an important part of the history. These include personal relationships, work pressures and problems related to the athlete's sporting activity. Identified problems require management as part of an overall care package.

Clinical measurement of headache

Headache is a complex multidimensional problem, which needs to be assessed using a range of different measurement tools including diaries and questionnaires. There are specific instruments that measure headache burden and severity.¹⁷ Primary measures of headache include attack frequency or headache days per month, which can be combined with headache severity and duration to form an index.¹⁸ Additional parameters include measures of disability and quality of life such as the headache disability inventory,¹⁹ as well as medication consumption, psychiatric symptoms, stress and coping, and treatment satisfaction. Simply measuring headache frequency, intensity and duration as an outcome measure may be problematic as people tend



Figure 21.2 Factors to consider in a headache evaluation

to either underestimate or overestimate their symptoms depending on how they feel at the time they are asked.¹⁷ Hence evaluating a range of headache parameters is likely to give a clearer, bias-free estimate of change. Figure 21.2 illustrates typical measurements taken during evaluation.

Examination

In all patients presenting with headache, a full neurological examination is required, and the skull, cervical spine and orofacial region must always be carefully examined. The examination should consist of some or all of the following components depending on the presence or absence of specific symptoms in the history:

- general appearance
- mental state
- speech
- skull examination
- cervical spine examination
- temporomandibular joint
- gait and stance
- pupils and fundi
- special senses (e.g. smell, vision, hearing)
- other cranial nerves
- motor system
- sensory system
- general examination.

PRIMARY HEADACHE

Primary headache includes migraine, tension-type headache, medication overuse headache, exercise-related headache and some types of post-traumatic headache.

In the general population, the 1-year prevalence rate for primary headache is 62%.²⁰ By survey, approximately 40% of migraine sufferers experience migraine precipitated by exercise, and 34% of Australian rules football players suffered migraine according to IHS criteria, a rate higher than population controls.⁷ In that study, almost 60% of subjects experienced regular headaches related to sport.⁴

Migraine

Previously primary headache, in particular migraine, was thought to be a vascular disorder, or due to cortical spreading depression. More recent evidence points to abnormality of brain function leading to a chain of events in the periphery.²¹⁻²³ In simple terms, migraine pain can be thought of as an altered perception of normality, such that normal sensory input is misinterpreted as pain.²⁴ This concept of the variable modulation of sensory input is explained in Lorimer Moseley's YouTube video 'Why things hurt' (<http://ow.ly/S9n0h>). (See also Chapters 5 and 6.)

The neuroanatomical basis for migraine is the trigeminocervical nucleus. The migraine pain process is likely to be a combination of direct factors, that is, activation of the trigeminal nociceptors, in concert with a reduction in the normal functioning of the centrally mediated endogenous pain control pathways that normally gate that pain.²⁵ Thus there are both central and peripheral mechanisms involved.

Migraine is broadly categorised into migraine with and without aura. An aura is a specific set of neurological symptoms that typically precede the headache which include visual disturbances (e.g. scotomas), paraesthesia, vertigo, hemiplegia and ophthalmoplegia. Although most people think of migraine as headache alone, the true migraine sufferer usually notices a spectrum of symptoms, including nausea, vomiting, diarrhoea and weight gain. The important point for sports clinicians is that there does not have to be an aura. The IHS criteria for the diagnosis of migraine without aura are shown in Table 21.2.³

Clinical features

The typical features of migraine are precipitation by a change in homeostasis such as tiredness, temperature, altitude, thirst, hunger or stress. Ensuing headache pain is described as sharp and intense, throbbing or beating in time with their pulse. Commonly, it begins in the temple or forehead on both sides. When it starts on one side, it may spread to the other side. Occasionally, the headache begins at the back of the head and moves forwards.

The common neurological accompaniments to migraine with aura are visual. Patients speak of bright or dark objects often to one side of the visual field. These objects may shine or flicker and typically move across the visual field. The visual symptoms usually last about 20 minutes

Table 21.2 Diagnostic criteria for migraine without aura³

- A. At least five attacks fulfilling criteria B–D below
- B. Headache attacks lasting 4–72 hours
- C. Headache has at least two of the following four characteristics:
1. unilateral location
 2. pulsating quality
 3. moderate or severe pain intensity
 4. aggravation by or causing avoidance of routine physical activity (e.g. walking or climbing stairs)
- D. During headache at least one of the following:
1. nausea and/or vomiting
 2. photophobia and phonophobia
- E. Not better accounted for by another diagnosis

and most often clear before the sensory, cognitive or headache symptoms begin. Sensory symptoms are usually described as tingling, pricking or pins and needles. These commonly commence in the face or fingers and gradually spread up the limb or over the same side of the body. Nausea, vomiting and dizziness are common during or after the attack.

Precipitating factors in migraine

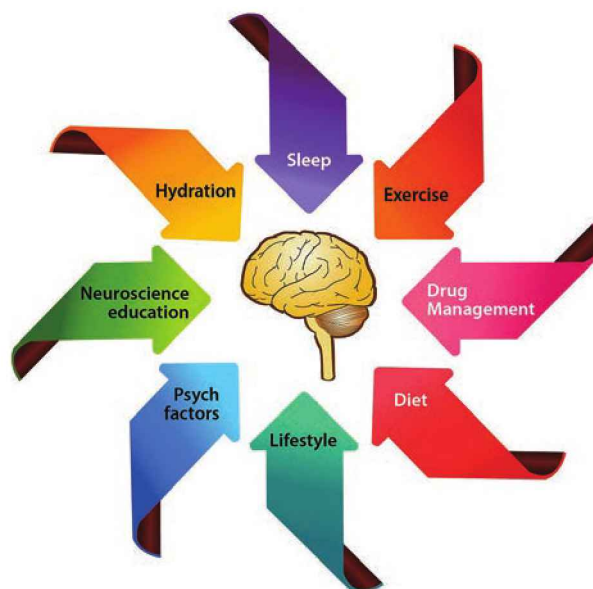
A number of precipitating factors in addition to a change in homeostasis are commonly found in association with migraine headaches. These are:

- endocrine changes (e.g. premenstrual or menstrual, oral contraceptive pills, pregnancy, puberty, menopause, hyperthyroidism)
- metabolic changes (e.g. anaemia, thyroid disease)
- infective causes (e.g. fever, rhinitis)
- change in activity including extreme exercise
- alcohol
- drugs (e.g. glyceryl trinitrate [nitroglycerine], nitrates, indomethacin).

Chocolate and cheese are anecdotally reported triggers of migraine, but this is not confirmed in trials.²⁶

Treatment

Although sleep and resting quietly in a dark room often helps alleviate a migraine attack, the primary method of active treatment is typically pharmacological. High-dose aspirin (ASA) (900 to 1200 mg), often combined with an antiemetic, is the drug of choice in the acute phase. Other acute agents such as oral, intranasal or parenteral triptans may be used as rescue medications if the initial therapy is unsuccessful. Determining the best drug option depends on a number of patient factors as well as the nature of the headache and timing of treatment. This is best done by a medical practitioner experienced in this area. A person experiencing frequent or disabling migraine episodes,

**Figure 21.3** Management of migraine

especially when he or she is losing considerable time off work or school, may find prophylactic therapy necessary.

The long-term management of migraine should be multifactorial (Fig. 21.3). An important part of this plan is to identify and avoid precipitating factors. Central to this is the patient's understanding of homeostasis, maintaining equilibrium through adequate diet, hydration and sleep, as well as reducing stress and anxiety.²⁷ Neuroscience education may also be valuable in this regard, and involves the careful explanation of pain mechanisms and central sensitisation. Other options include traditional herbal remedies such as 'feverfew'.

Exercise can induce and exacerbate migraine, but exercise can also activate a pain modulatory mechanism, ideal in the management of migraine.²⁸ The prophylactic effects of exercise in migraine were shown in a randomised controlled

trial.²⁹ Intervention comprised three arms: home relaxation; controlled aerobic exercise on a static bike; or the oral medication topiramate. The study included an extended follow-up period. Exercise was equally effective as medication in reducing episodes of migraine. A second pilot study evaluated the effects of an 8-week running program and also found positive benefits compared to a control group.¹⁵

It is critical in the management of migraine and other forms of headache that the use of repeated doses of analgesia alone be avoided. One of the consequences of the overuse of analgesic medication is medication overuse headache, which becomes a self-generating headache. The underlying mechanisms of medication overuse headache appear similar to migraine with alteration in cortical neuronal excitability and central sensitisation. The diagnosis is based on the presence of chronic headache on 15 days per month for 3 months, with simple analgesic medication taken on 15 days per month.³ The condition can be managed with variable success with a structured management program.³⁰ For this reason, the use of simple analgesics in headache treatment should be limited to a maximum of 3 days per week. Drug management should be directed at the cause of the problem, not simply short-term pain relief.

Primary exercise headache

Primary exercise headache is precipitated by exercise in the absence of any intracranial disorder. It has been reported in association with weightlifting, cycling, running and other sporting activities. The IHS criteria include that the headache:

- is specifically brought on by sustained strenuous physical exercise
- has a pulsating quality that lasts from 5 minutes to 48 hours
- is prevented by avoiding excessive exertion
- occurs particularly in hot weather or at high altitude.³

The onset of the headache can be with straining and Valsalva manoeuvres such as those seen in weightlifting and competitive swimming. The prevalence may be high, as headache consistent with primary exercise headache was reported in 26% of 4000 competitors in a tough cycling event in Holland, and was particularly associated with extreme exertion, low fluid intake and warm weather.³¹ In the Vaga study in Norway, 12% of a sample of 1646 people were reported to have primary exercise headache.³² Patients with this form of headache have significantly higher prevalence of internal jugular venous valve incompetence, which suggests that intracranial venous congestion caused by retrograde jugular venous flow may play a role in the pathophysiology of this disorder.³³ The major differential diagnoses are subarachnoid haemorrhage, cervical arterial dissection and reversible

cerebral vasoconstriction syndrome, which need to be excluded by the appropriate investigations.

It has been postulated that primary exercise headache is due to dilatation of the pain-sensitive venous sinuses at the base of the brain as a result of increased cerebral arterial pressure due to exertion.³⁴ In weight lifters, systolic blood pressure may reach levels above 400 mmHg and diastolic pressures above 300 mmHg with maximal lifts.

The management of this condition involves either avoiding the precipitating activity or drug treatment, for example, indomethacin (25 mg three times a day). In practice, the headaches tend to recur over weeks to months and then slowly resolve, although in some cases they may be lifelong.

Exertion may also be a triggering factor in migraine. In this case, the presentation shows the typical pattern of migraine with exertion as the precipitating factor. Most patients with this condition describe the migraine beginning immediately after exercise, more frequently when the exercise has been vigorous. Exertional migraine is often severe and like primary exercise headache may be worse in hot weather. Modulation of exercise programs including adequate warm up, particularly prior to intense periods of exercise or before a game, is important. Where this does not help, migraine prophylactic medication may be required.

SECONDARY HEADACHE

Cervicogenic headache

A large survey of the general population revealed that people who suffer from headache seek physical treatment from physiotherapists and other complementary medical practitioners.³⁵ These people believe that their headache may be a referred phenomenon from a disorder affecting structures in their cervical spine. While not all patients with headache respond to these varied treatments, many do and there is now substantial evidence of a distinct subgroup of headache, termed cervicogenic headache, where treatment to the cervical spine has a substantial lasting effect.³⁶

Mechanism

The underlying pathophysiology of cervicogenic headache is one of convergence, whereby afferents from the upper three cervical nerve roots converge in the same brain region as trigeminal afferents.³⁷ Hence, in the presence of sensitisation of this complex, cervical afferent input is misinterpreted, and as well as perceiving neck pain, headache is also felt. Evidence of such sensitisation in cervicogenic headache was shown in a study comparing trigeminocervical sensory processing in patients with cervical facet joint pain with and without headache.³⁸

Sensitisation of the trigeminocervical nucleus is an important factor in many different forms of headache.³⁸⁻⁴⁰ To explain this, it has been postulated that primary headaches form a spectrum, with shared common

pathophysiological mechanisms,⁴¹ and with cervicogenic headache one part of this spectrum,⁴² despite being a secondary headache. Notwithstanding this shared pathophysiology among different headache forms, it is apparent from systematic reviews that different headache forms respond to different forms of intervention; for example, in the long-term, migraine does not respond to physical intervention.⁴³ Knowledge of the underlying mechanisms of headache may provide insight into potential management and will be discussed later.

Clinical features

A thorough history and comprehensive examination is required.

History

The subjective features of cervicogenic headache contrast to those reported by people with migraine and other headache forms. Table 21.3 identifies the characteristics

of cervicogenic headache and Table 21.1 illustrates the difference in presentation between migraine and cervicogenic headache based on a survey of all people with chronic headache evaluated in a small town in Norway.^{9, 10} Despite this, it can still be difficult to distinguish migraine from cervicogenic headache and an incorrect diagnosis is made in up to 50% of cases.^{44, 45}

A good example might be a cyclist who develops headache during training after a specific time period on his or her bike. This could be exercise-induced migraine or cervicogenic headache from neck stress as a consequence of an incorrectly adjusted bike or poor posture and associated altered muscle function. In this case, the physical examination becomes increasingly important in differential diagnosis. A further difficulty is that patients more often than not present with multiple rather than single headache forms, making definitive diagnosis more challenging. Amiri et al. reported that 55% of subjects in a sample of 196 people had multiple forms of headache, with up to four different types.⁴⁶

Table 21.3 Classification of cervicogenic headache

	International Headache Study Group ⁴⁷	International Headache Society ³
Symptoms	• Unilateral headache without side shift • Ipsilateral neck, shoulder and arm pain of a rather vague, non-radicular nature • Pain episodes of varying duration or fluctuating, continuous pain • Moderate, non-excruciating pain, usually of a non-throbbing nature • Pain starting in the neck, eventually spreading to head, where the maximum pain is often located • Pain triggered by neck movement and/or sustained awkward position • Sustained neck trauma prior to the onset • Autonomic symptoms and signs (e.g. nausea, vomiting, dizziness, photo- and phonophobia, blurred vision)	A. Any headache fulfilling criterion C
Physical examination	• Reduced cervical spine range of motion • Symptoms on palpation of the cranium or neck • Anaesthetic blockades abolish the pain transiently	B. Clinical, laboratory and/or imaging evidence of a neck disorder, known to be able to cause headache C. Evidence of causation demonstrated by at least two of the following: 1. Headache has developed in temporal relation to the onset of the neck disorder 2. Headache has significantly improved in parallel with improvement in the neck disorder 3. Cervical range of motion is reduced and headache made worse by provocative manoeuvres 4. Headache is abolished following cervical diagnostic anaesthetic blockade
Other		D. Not better accounted for by another headache diagnosis

Examination

Any structure innervated by the upper three cervical nerve roots is a potential contributing factor to sensitisation of the trigeminocervical nucleus and a factor in cervicogenic headache, hence a thorough clinical examination is required of these structures including the cervical articulations C0-1, C1-2 and C2-3, as well as cervical myofascial and neural structures. Caution is required as abnormalities found do not necessarily indicate the involvement of the cervical spine in headache and need to be considered in a diagnostic framework. For example, cervical movement impairments⁴⁸ and abnormal neurodynamic tests⁴⁹ have been reported in people with migraine.

There are a number of classification systems described to aid in identifying cervicogenic headache. These include those published by the IHS³ and the International Headache Study Group¹³ and are shown in Tables 21.1 and 21.3. Using the criteria in Table 21.1, it is possible to gain a level of confidence regarding the diagnosis of cervicogenic headache, with five criteria required for diagnosis.⁴⁷ Successful management from treatment directed to the cervical spine would be confirmatory evidence based on the IHS criteria.³ In the absence of these criteria, further clinical evaluation should be undertaken.

PRACTICE PEARL

Examination of the patient with suspected cervicogenic headache involves a systematic examination of the articular, myofascial and neural structures of the cervical and orofacial region, including temporomandibular joints.

As with any musculoskeletal examination, one of the aims of the examination is to reproduce the patient's symptoms, but this by itself does not indicate a musculoskeletal cause for pain. In a recent study, headache was reproduced on palpation of the cervical spine in all but one patient with migraine and tension-type headache.⁴² If headache can be provoked and, importantly, then relieved by manual techniques to the cervical spine then there is greater confidence of a cervical musculoskeletal cause and the potential for such treatment to be effective in the management.

Articular function can be determined by examining range of motion of the cervical spine, either in single cardinal planes or combined. Combining movements



Figure 21.4 The cervical flexion–rotation test

increases the potential to identify movement impairment at the segmental levels between the occiput and C3. One example is the flexion-rotation test (Fig. 21.4), which can be used to isolate impairment of movement in the upper cervical spine. The neck is placed in end-range flexion and range of rotation to each side recorded. Normal range is 44° to each side.⁵⁰ The minimum cut-off value for a positive test is approximately 33° and there is a close relationship between headache severity and restricted range of movement (ROM) on a positive test.⁵¹⁻⁵⁴ This test has also been shown to be helpful, with good diagnostic accuracy, in distinguishing people with migraine from those with cervicogenic headache.⁵²

A positive test is not universal in cervicogenic headache, but is found in as many as 72%.⁵³ While the test has been shown to be a valid marker of C1-2 segmental movement in normal people,⁵⁵ potentially unilateral altered suboccipital muscle tone (obliquus capitis inferior) associated with any painful articulation above C3, may give a positive flexion-rotation test. With further careful examination of the articular structures by palpation and segmental motion tests, it is possible to determine the spinal level of involvement.⁵⁶

Temporomandibular disorder (TMD) is a common accompaniment to different headache forms, not specifically cervicogenic headache.⁵⁷ Evidence for the relationship between impairment of the cervical spine and TMD is shown in the significant correlation ($r = 0.82$) between jaw disability and neck disability.⁵⁸ Upper cervical movement impairment identified by the flexion-rotation test is found in people with TMD,⁵⁹ and in people with TMD and cervicogenic headache.⁶⁰ Careful examination of the temporomandibular joints and

orofacial muscles should be performed according to recommended guidelines.⁶¹

Identification of impairment of cervical motor control and myofascial system is important in cervicogenic headache evaluation,⁶²⁻⁶⁵ as illustrated by evidence of long-term improvement in cervicogenic headache symptoms following retraining cervical and scapular motor control.⁶⁶ Impairment includes loss of postural alignment (forward head posture) and neuromuscular control as well as muscle weakness, endurance and extensibility.⁶⁷

Impairments of craniocervical flexion control are some of the defining features of neck-related headache.^{62, 63, 68} This test evaluates the deep neck flexor muscles, which typically become substituted by the superficial flexors sternocleidomastoid, hyoids and platysma. Similar impairments in craniocervical flexion control have not been found in migraine or tension-type headache according to one study,⁶² but were in tension-type headache in another.⁶⁹ Other changes to the muscle system in cervicogenic headache include evidence of muscle tightness⁶⁵ and trigger points.⁷⁰ See Chapter 23 for more details of evaluation of cervical muscle control.

Sensitisation of the upper cervical neural tissue is found in approximately 10% of cases with cervicogenic headache, less commonly when compared to the cervicobrachial region.^{65, 71} Identification is important, as the presence of such sensitisation is likely to lead to treatment failure unless addressed. Typically, the patient will adopt an antalgic forward head posture with retraction and upper cervical flexion will be provocative and limited in range. Suboccipital muscle tone may be raised as a protective measure. Neurodynamic tests (Chapter 15), such as in Figure 21.5 involving upper cervical spine flexion with the arms in a neural provocative position, should reproduce head pain. The upper cervical neural tissue (e.g. greater occipital nerve) is likely to be sensitive to palpation in positive cases (Fig. 21.6).

In this section, various aspects of examination have been described to aid in the identification of cervicogenic headache. While the individual items of assessment may be of importance, consideration of multiple physical examination criteria is likely to increase the accuracy of diagnosis. This is supported by one study which revealed that collectively, impairment in motor control identified by the craniocervical flexion test and restricted neck movement, in association with evidence on manual examination of upper cervical joint dysfunction had 100% sensitivity and 94% specificity to identify cervicogenic headache.⁶²



Figure 21.5 Upper cervical neurodynamic test involving upper cervical spine flexion with the arms in various neural provocative positions, with an aim to reproduce head pain



Figure 21.6 Palpation of the greater occipital nerve

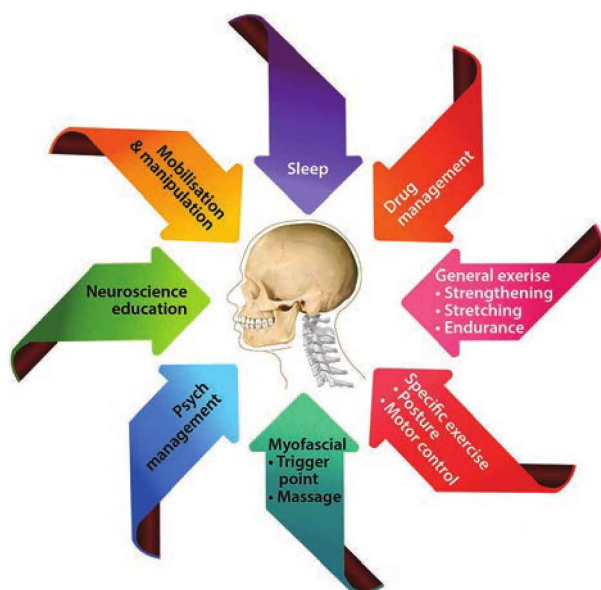


Figure 21.7 Management of cervicogenic headache

Treatment

The clinician's confidence in diagnosis of cervicogenic headache, and thus potential for response to physical interventions, are improved by combining information from all aspects of the examination, as currently there is no consensus as to which examination criteria predict success for cervicogenic headache.^{72, 73} Hence treatment, as illustrated in Figure 21.7, should ideally target all factors identified during the examination as contributing to the athlete's problem. An important factor associated with recovery from neck pain is sleep quality,⁷⁴ hence physical factors (articular, neural and muscle function) are not the only ones to consider in management. Since cervicogenic headache is associated with sensitisation of the trigeminocervical nucleus, factors that might reduce this, such as neuroscience education and lifestyle management, as well as reducing stress and anxiety, should also be considered.

Evidence for a multisystem approach to management was shown in a multicentre randomised controlled trial.⁶⁶ While manipulation/mobilisation and, separately, motor control re-education were found to be effective in the long-term, combining both these approaches was found to be more effective.

There is evidence for a range of different manual therapy techniques in management of cervicogenic headache. Manipulation of the cervical spine may be effective,⁷⁵ but is potentially more hazardous and has been found to be no more effective than more gentle, potentially less dangerous forms of treatment including joint mobilisation or Mulligan mobilisation with movement.⁷⁶ Joint



Figure 21.8 Passive upper cervical rotation mobilisation technique

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mobilisation might include Maitland passive accessory mobilisation or physiological mobilisation techniques in appropriate cervical positions (based on assessment) to stretch the impaired joints to restore normal movement (Fig. 21.8). Such techniques should include exercise to maintain improved mobility.

Mobilisation with movement is a combination of pain-free passive accessory joint glide applied together with active movement (Fig. 21.9) and can be used to improve segmental joint mobility at any level in the cervical spine.⁷⁷ The aim of this approach is to restore normal segmental movement in a painless way by encouraging normal biomechanics. For example, a cyclist may develop cervicogenic headache due to abnormal movement at the C2-3 segment, which might provoke headache during prolonged cycling activity. Improving C2-3 segmental motion with a mobilisation with movement technique together with a self-mobilisation home exercise, postural control exercise and muscle re-education typically eliminates the problem.

Another common clinical situation in athletes with chronic cervicogenic headache is a positive flexion-rotation test (Fig 21.4). This usually responds in a rapid and sustained way to a simple Mulligan self-mobilisation technique (Fig. 21.10) directed to the upper cervical spine.⁷⁸ In this case, treatment is aimed at the impaired cervical articulations in an attempt to increase mobility specifically in the upper



Figure 21.9 Mobilisation with movement into flexion at C2–3
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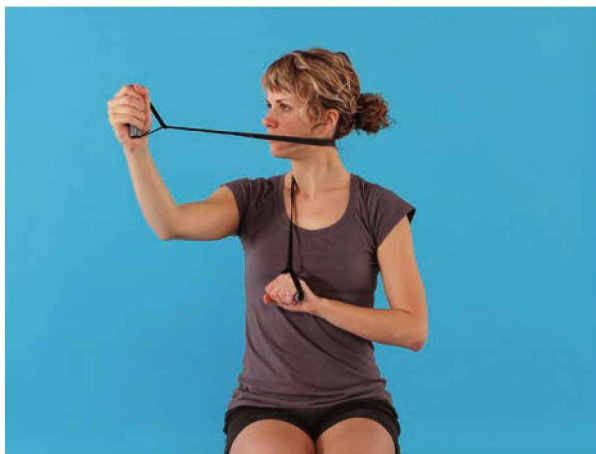


Figure 21.10 Mulligan self-mobilisation for upper cervical rotation impairment identified by the flexion–rotation test

cervical spine. The positive sustained long-term improvement following this intervention goes some way to validate the use of mobilisation in cervicogenic headache.⁷⁸

Dosage of mobilisation and manipulation is important. For example, a single session of manipulation is not effective in the long term,⁷⁹ but eight sessions are effective, as are 16.⁷⁵ Specific trigger point and other soft tissue techniques may be used if these have been found to be involved in the disorder. There is some evidence to show that simple soft tissue therapy techniques can improve the flexion-rotation test, thus helping headache symptoms.⁸⁰



Figure 21.11 Tape to improve scapula control

PRACTICE PEARL

A comprehensive exercise program should include motor control exercises including postural control, particularly in the functional complaint activity/position.

For example, cervical and scapular postural control needs to be addressed while riding a bike in the patient with cycling-induced cervicogenic headache, but to achieve this may also require attention to lumbopelvic posture and bike set-up. Postural faults may develop through repetitive sporting activity. For example, an increased thoracic kyphosis, cervical protrusion and scapula anterior tilt and protraction may develop in a swimmer, cyclist or body builder through uncorrected muscle activity. This position may abnormally stress the upper cervical spine, inducing headache. Correcting these faults may require joint mobilisation as well as postural and motor control re-education. Sports tape may be used to facilitate postural correction of the scapula (Fig. 21.11) until the athlete learns to correct his or her posture independently.

Exercise may also include stretching of any muscles found to be short, together with strengthening of the shoulder girdle and neck muscles. Important muscles for stretching include pectoralis minor, upper trapezius, levator scapula and suboccipitals, among others.⁶⁵ Neck strengthening combined with stretching exercises were found to be more effective compared to endurance training

Table 21.4 The IHS criteria for the diagnosis of headache attributed to trauma or injury to the head and/or neck (IHS 5.1 and 5.2)³

A. Any headache fulfilling C and D
B. Traumatic injury to the head
C. Headache is reported <7 days after:
1. injury to the head
2. regaining consciousness following head injury
3. discontinuing medication that impairs ability to sense headache following injury to the head
D. Headache disappears within 3 months after trauma (acute) or headache persists for greater than 3 months (persistent)

or stretching alone in the management of cervicogenic headache.⁸¹

Recent evidence suggests a close interaction between the upper cervical spine, temporomandibular dysfunction and cervicogenic headache.^{59, 82} This might explain why combining manual therapy techniques to the temporomandibular joint and orofascial muscles with cervical manual therapy was found to be more effective in the management of cervicogenic headache than cervical manual therapy alone.⁸³ The clinician should consider incorporating management of temporomandibular dysfunction when this is evident in examination.

Post-traumatic headache

Trauma to the head and neck in sport may lead to the development of headache. The initiating traumatic event may not necessarily be severe and could involve repetitive minor blows as in boxing. The IHS diagnostic criteria for post-traumatic headache are shown in Table 21.4.³

Most often these forms of headache resemble cervicogenic headache, tension-type or migraine. Not surprisingly, trauma to the head affects the neck and could lead to cervicogenic headache. Indeed, the presence of neck pain appears important in post-traumatic headache sufferers without premorbid headache.⁸⁴ Hence the diagnosis is made on the close temporal relationship between the traumatic event and the onset of headache, but should also require careful examination of the cervical spine and the potential for cervicogenic headache.

The presence of migraine-like headache after concussion correlates with more severe injuries and identifies athletes at risk of prolonged recovery to baseline.⁸⁵ Headache may occur in isolation or may be part of a range of symptoms including dizziness, sleep disturbance, balance disturbance and loss of cognition. This is dealt with in more detail in Chapter 20 on concussion.

Evidence for the management of post-traumatic headache is not well established. Generally, treatment should be provided according to the category the particular headache most resembles.⁴ The clinician should be aware

of the potential for medication overuse headache, a known factor in the development of persistent symptoms, unless analgesic medication is carefully monitored.⁸⁶

Post-traumatic migraine

Headache may follow repetitive relatively minor traumatic events to the head and neck. This may be seen in sports such as soccer, where repetitive heading of the ball gives rise to the term 'footballer's migraine'.^{87, 88} Boxing and similar sports also lead to problems like these. Even mild head trauma can induce migraine. One particular syndrome that is recognised in the setting of minor head blows is migrainous cortical blindness. This disturbing condition often raises fear of serious cerebral injury but tends to resolve over 1–2 hours.

External compression headache

External compression headache (IHS 4.2), formerly known as 'swim goggle headache,' presents with pain in the facial and temporal areas produced from wearing excessively tight face masks or swimming goggles. It is commonly seen in swimmers and divers. In divers, this may be referred to as 'mask squeeze', and is seen on descent to depth as the effects of pressure reduce the air space inside the mask. The aetiology is believed to be due to continuous stimulation of cutaneous nerves by the application of pressure.

High-altitude headache

High-altitude headache (IHS 10.1.1) is a well-recognised accompaniment of acute mountain sickness (although it can occur on its own), which occurs within 24 hours of ascent to altitudes above 3000 m.⁸⁹ The headaches are vascular in nature and are seen in unacclimatised individuals. Typically, these are associated with other physiological effects of altitude or may be an early manifestation of acute mountain sickness. The treatment is to descend to lower altitude, although pharmacological interventions such as ibuprofen, acetaminophen/paracetamol and acetazolamide may be used.

Hypercapnia headache

Headache in divers, while uncommon and generally benign, can occasionally signify serious consequences of hyperbaric exposure such as arterial gas embolism, decompression sickness, and otic or paranasal sinus barotrauma. Inadequate ventilation of compressed gases can lead to carbon dioxide accumulation, cerebral vasodilatation and headache.⁹⁰

Since carbon dioxide (CO₂) relaxes cerebrovascular smooth muscle, hypercapnia leads to cerebral vasodilatation and increased intracranial pressure. Carbon dioxide can accumulate insidiously in the diver who intentionally holds the breath intermittently (skip breathing) in a mistaken attempt to conserve air, or takes shallow breaths to minimise buoyancy variations in the narrow passages of a wreck or cave. The diver may hypoventilate unintentionally if a tight wetsuit or buoyancy compensator jacket restricts chest wall expansion, or if ventilation is inadequate in response

to physical exertion when swimming against a strong current.

The headache associated with CO₂ retention develops gradually and affects the bifrontal, bitemporal or bioccipital head regions. Throbbing pain ranging from mild to severe in intensity typically lasts from 10 to 30 minutes, yet in some may persist for several hours along with nausea and malaise. Onset accompanies a slow buildup or may follow a reduction in sharply rising CO₂ tension. Pain typically worsens during the decompression phase of the dive or upon surfacing, possibly because of the disparity between increased intracranial pressure and more rapidly decreasing ambient pressure. Non-steroidal anti-inflammatory and ergotamine preparations have been reported to be ineffective.⁹⁰

Other types of headache encountered in divers include exertional headache, cold stimulus headache, migraine, tension-type headache, acute traumatic headache, cervicogenic headache, carbon monoxide poisoning headache and headache associated with envenomation.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

